

Artificial Intelligence Readiness Scale

Extending UTAUT2 for Enterprise AI Adoption



Fabio Correa

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Dissertation Committee

Dr. Karina Kasztelnik

Committee Chair

Dr. Jerome Jones

Committee Member

Dr. Donna Day

Committee Member

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A Dissertation Presented in Partial Fulfillment of the Requirements for the Degree Doctor of Business Administration

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This presentation introduces the Artificial Intelligence Readiness Scale (AIRS)—a diagnostic instrument developed to help organizations understand why individuals adopt or resist AI tools. The presentation covers the problem motivation, theoretical framework, methodology, key findings, and contributions to both theory and practice.

Estimated duration: 20–25 minutes plus Q&A

The AI Adoption Paradox

Organizations invest heavily in AI but lack tools to understand *why* individuals adopt or resist. McKinsey's 2025 State of AI Global Survey reveals a striking disconnect between widespread adoption and value realization.

88%

Organizations Using AI

Report AI adoption in at least one business function (McKinsey, 2025)

~6%

AI High Performers

Qualify as high performers with meaningful EBIT impact (McKinsey, 2025)

90-95%

Pilot Failure Rate

Generative AI pilots fail to scale (MIT Media Lab NANDA Initiative, 2025)

The Gap: Organizations excel at deploying AI technology but lack understanding of the human factors—why individuals adopt or resist AI tools. This research addresses that gap.

Three Interconnected Gaps

Why This Research Matters

01

Theoretical Gap

UTAUT2 validated on mobile apps, internet banking, consumer tech—not AI (Venkatesh et al., 2012). Venkatesh (2021) identified nine unique AI research challenges unaddressed in existing frameworks, including trust in opaque systems, ethical concerns, and algorithmic decision-making.

02

Measurement Gap

No psychometrically validated AI adoption instrument exists. Organizations rely on ad hoc surveys or general technology readiness scales not designed for AI-specific characteristics.

03

Practice Gap

Without valid measurement, organizations cannot diagnose why AI initiatives fail or identify which employees will adopt versus resist. This contributes directly to the 90–95% pilot failure rate.

This research addresses all three gaps simultaneously.

Research Questions

Primary Research Question

How can UTAUT2 be extended with AI-specific constructs to better predict behavioral intention to adopt AI tools in professional and academic contexts?

Secondary Research Questions

RQ	Focus	Answer
RQ1	Factor structure	8 factors, 16 items
RQ2	Measurement invariance	Configural invariance achieved
RQ3	Strongest predictors	PV > HM > SI
RQ4	AI Trust extension	Marginal effect (p = .064)
RQ5	Moderators	Experience & population moderate HM

Theoretical Framework

UTAUT2 + AI Trust Extension

Why UTAUT2?

Explanatory power: UTAUT2 explains up to 74% of variance in behavioral intention versus ~40% for TAM—nearly double the predictive accuracy.

Theoretical integration: Synthesizes eight prior models (TRA, TPB, TAM, Motivational Model, etc.), representing 30 years of accumulated research.

Consumer-relevant constructs: Includes Hedonic Motivation, Price Value, and Habit—critical for AI adoption contexts.

Current gold standard: Building on the strongest established foundation ensures meaningful contribution to the field.

AI-Specific Extension: Trust

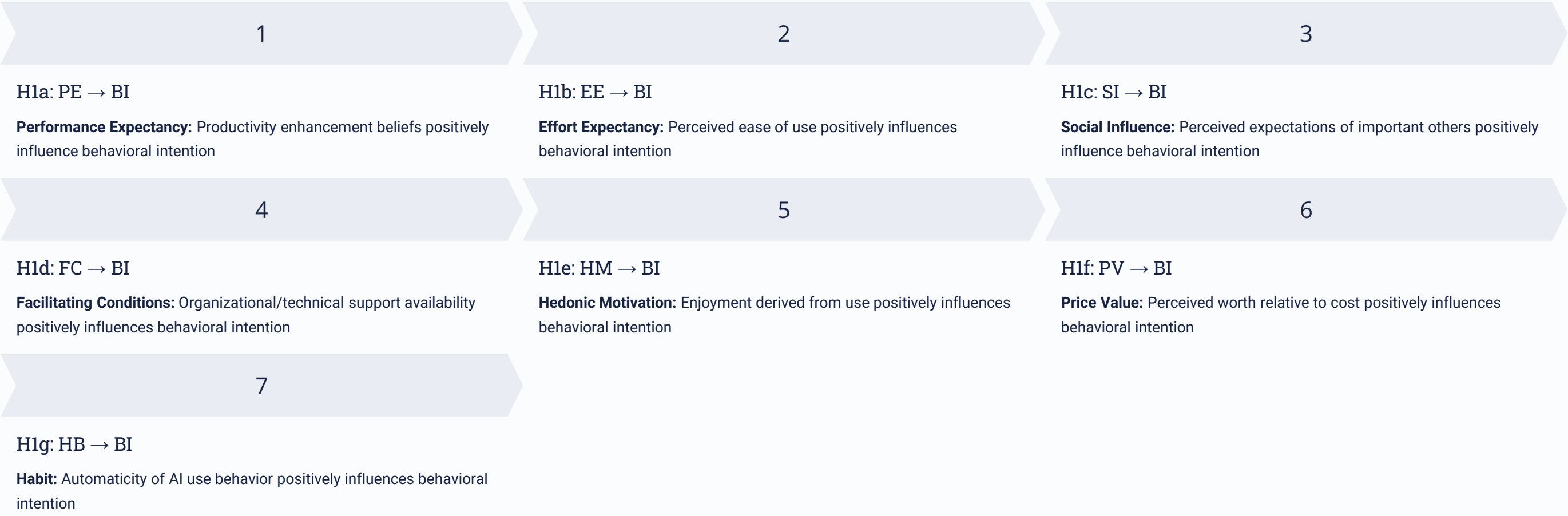
AI is characterized by opacity, probabilistic reasoning, and autonomous decision-making. Glikson and Woolley (2020) established in *Academy of Management Annals* that trust in AI differs fundamentally from trust in traditional technology.

AI systems cannot be trusted the same way as deterministic tools like spreadsheets. This unique characteristic requires theoretical extension beyond UTAUT2's original scope.

Core Hypotheses

UTAUT2 Core Hypotheses (H1a–H1g)

Based on Venkatesh et al. (2012)



AI Extension & Moderators

Novel Hypotheses (H2–H4)

H2: AI Trust Extension

AI Trust → BI (positive)

Tests whether AI Trust predicts adoption beyond UTAUT2's established constructs

Theoretical basis: Glikson & Woolley (2020)

H3: Experience Moderation

Experience moderates paths

Tests whether experience level moderates predictor-intention relationships

Theoretical basis: Venkatesh et al. (2003)

H4: Population Moderation

Population moderates paths

Tests whether population type (students vs. professionals) moderates predictor-intention relationships

Theoretical basis: Novel extension

Research Design

Methodology Overview

Design

Cross-sectional, quantitative

Cross-sectional, quantitative design with N = 523 participants from both academic and professional contexts. Initial instrument contained 28 items across 12 constructs; refined through rigorous psychometric analysis to 16 items across 8 factors.

Sample

N = 523 (students + professionals)

Analysis pipeline follows instrument development best practices:

1. **EFA** (Exploratory Factor Analysis): Structure discovery
2. **CFA** (Confirmatory Factor Analysis): Structure validation
3. **SEM** (Structural Equation Modeling): Hypothesis testing

Instrument

28 items → 16 items (validated)

Analysis

EFA → CFA → SEM

Psychometric Validation

Model Fit Results

Psychometric results demonstrate strong model fit, meeting publication-quality standards across all major indices:

Metric	Result	Threshold	Reference
CFI	.975	> .95	Hu & Bentler (1999)
TLI	.960	> .95	Hu & Bentler (1999)
RMSEA	.065	< .08	Browne & Cudeck (1993)
All α	> .74	> .70	Nunnally (1978)

- **CFI = .975:** Exceeds .95 excellent threshold
- **TLI = .960:** Exceeds .95 excellent threshold
- **RMSEA = .065:** Below .08 acceptable cutoff
- **All α > .74:** All reliability coefficients exceed .70 threshold

The instrument meets publication-quality psychometric standards.

Sample Demographics

Participant Characteristics (N = 523)

Diverse sample with rigorous quality controls enables robust cross-population analysis and moderation testing:



Population Split

41% academic / 59% professional—
enables cross-population invariance
testing



Education Range

34% some college, 28% bachelor's,
18% master's—high school to doctoral
degrees represented



Experience Variation

37% with 11+ years / 19% with <1
year—enables experience moderation
testing



Leadership Representation

24% in supervisory/executive roles
(managers + executives)



Disability Representation

13% identify as having a disability—
enables accessibility analysis

Recruitment protocol: Centiment professional research panel with blinded recruitment (topic undisclosed until consent form), multiple embedded attention checks, and exclusion of responses failing quality controls—reduces self-selection bias and ensures data quality (Meade & Craig, 2012).

Key Finding #1: What Drives AI Adoption

Structural Model Results (N = 523)

Structural model results showing path coefficients for all predictors of Behavioral Intention. Model explains **85.2% of variance**—exceptional predictive accuracy.

Predictor	β	p	Result
Price Value	.505	<.001	✓ Strongest
Hedonic Motivation	.217	.014	✓ Supported
Social Influence	.136	.024	✓ Supported
AI Trust	.106	.064	○ Marginal
Performance Expectancy	−.028	.791	✗ Not Supported
Effort Expectancy	−.008	.875	✗ Not Supported
Facilitating Conditions	.059	.338	✗ Not Supported
Habit	.023	.631	✗ Not Supported

Model R² = .852 (85.2% variance explained)

Key Finding #1: The Value Paradigm Shift

What Works (Significant)

- **Price Value ($\beta = .505$):** Strongest—cost-benefit evaluation dominates adoption decisions
- **Hedonic Motivation ($\beta = .217$):** Second—enjoyment drives sustained adoption
- **Social Influence ($\beta = .136$):** Third—peer influence matters significantly
- **AI Trust ($\beta = .106$):** Marginal but diagnostic value retained

What Fails (Not Significant)

- Performance Expectancy (productivity gains)
- Effort Expectancy (ease of use)
- Facilitating Conditions (support/training)
- Habit (prior experience)

This pattern represents a **fundamental departure from 30 years of technology acceptance research**. Traditional UTAUT2 workhorses fail for AI adoption while value-focused and enjoyment-focused factors dominate.

Key Finding #2: Traditional Drivers Fail

Why 30 Years of Research Doesn't Apply

Contrasts with Blut et al. (2022) meta-analysis of 737,112 users where Performance Expectancy was strongest predictor ($\rho = .60$)

Performance Expectancy

"Will this make me productive?"

NOT Significant

Productivity gains don't predict AI adoption

Effort Expectancy

"Is this easy to use?"

NOT Significant

Ease of use doesn't predict AI adoption

Facilitating Conditions

"Do I have support?"

NOT Significant

Training/support doesn't predict AI adoption

Habit

"Have I used this before?"

NOT Significant

Prior experience doesn't predict AI adoption

Performance Expectancy has been the dominant predictor for 30 years across PCs, email, smartphones, and enterprise software. **For AI: not significant.** Same pattern for Effort Expectancy, Facilitating Conditions, and Habit.

Key Finding #2: The Paradigm Shift

Traditional Tech vs. AI Adoption

1

Traditional Tech

"Will this make me productive?"

Feature-focused evaluation

Training solves adoption barriers

2

AI Adoption

"Is this worth my total investment?"

Value-focused evaluation

Trust enables adoption decisions

📌 **Critical Insight:** Organizations approaching AI adoption with traditional playbooks will fail. AI requires fundamentally different change management strategies—demonstrating value rather than capabilities, and building trust before exposing features.

Key Finding #3: User Segmentation

Four Distinct AI Adoption Profiles

K-means cluster analysis (k = 4, silhouette = .31)

Four distinct user segments identified via cluster analysis on standardized UTAUT construct scores. Each segment has unique characteristics requiring different intervention strategies.

Segment	%	Trust	Anxiety	Intent
AI Enthusiasts	16%	+1.37σ	−0.86σ	+1.32σ
Cautious Adopters	30%	+0.86σ	+0.84σ	+0.88σ
Moderate Users	37%	+0.01σ	+0.42σ	−0.07σ
Anxious Avoiders	17%	−1.49σ	+0.76σ	−1.53σ

Cluster analysis based on standardized UTAUT construct scores, adapting Rogers (2003) innovation adoption categories.

Key Finding #3: Segment Strategies

Targeted Interventions by Segment

	AI Enthusiasts (16%) Strategy: Champions Give early access to new tools, enable evangelization, leverage as internal advocates. These users will drive organic adoption through peer influence.
	Cautious Adopters (30%) Strategy: Proof Points Provide case studies, ROI calculators, low-commitment pilots. These users need evidence before committing—show them the value.
	Moderate Users (37%) Strategy: Reduce Friction Demonstrate peer adoption, leverage social influence, make adoption the path of least resistance. These users follow the crowd.
	Anxious Avoiders (17%) Strategy: Trust-First Address anxiety before features; trust-building required. Feature demonstrations are useless until trust is established.

AIRS-16 scores enable segment identification for targeted interventions instead of one-size-fits-all training programs. Each segment requires fundamentally different approaches.

Moderator #1: Experience

Experience Moderates Hedonic Motivation

Interaction	β	p
HM × Experience	.136	.009
PE × Experience	.112	.055

Interpretation: The HM × Experience interaction is significant (p = .009). Experienced professionals (4+ years) weight enjoyment *more* heavily in adoption decisions, not less.

This is a counterintuitive finding—we expected experienced users to be more pragmatic. Possible explanation: experienced professionals have learned that if a tool isn't enjoyable, sustained use is unlikely. Enjoyment predicts long-term adoption commitment.

Moderator #2: Population

Population Differences

Dramatic difference in Hedonic Motivation path across populations reveals fundamental cultural differences in how academics and professionals evaluate AI tools.

Path	Academic β	Professional β	$\Delta\beta$	p
HM $\rightarrow$ BI	.449	-.301	-.750	.041

Academics

β = .449 (highly positive)

Enjoyment drives adoption—academic contexts value tool enjoyment and exploration

Professionals

β = -.301 (negative)

Enjoyment may signal unseriousness—professional contexts may have cultural norms that work tools shouldn't be "fun"

Interpretation: This finding has direct implications for audience-specific AI messaging strategies. What works in academic contexts may backfire in professional contexts.

The AIRS-16 Instrument

Validated 8-Factor, 16-Item Diagnostic Tool

Factor	Items	α	Purpose
Performance Expectancy	2	.80	Productivity beliefs
Effort Expectancy	2	.86	Ease of use
Social Influence	2	.75	Peer/org influence
Facilitating Conditions	2	.74	Support availability
Hedonic Motivation	2	.86	Enjoyment
Price Value	2	.88	Worth vs. cost
Habit	2	.91	Behavioral patterns
AI Trust	2	.89	AI-specific trust

Administration: ~3 minutes | **Diagnostic output:** Identifies specific barriers

Organizations can administer the assessment, score it, and identify specific barriers: Low Price Value → Value demonstration intervention; Low Social Influence → Champion recruitment; Low Trust → Trust-building intervention first. AIRS-16 moves organizations from guessing about adoption barriers to diagnosing them with validated measurement.

Theoretical Contributions

Advancing Technology Acceptance Theory

01

Value-Dominant Model

Price Value ($\beta = .505$) dominates Performance Expectancy—challenges established TAM/UTAUT findings (Davis, 1989; Venkatesh et al., 2003, 2012). AI is evaluated through cost-benefit lens, not productivity lens. This reverses 30 years of technology acceptance findings.

03

AI Trust as Diagnostic Factor

Marginal predictive effect ($p = .064$) but essential diagnostic value. Enables targeted intervention design. Retained in instrument for diagnostic value—enables identification of trust deficits.

02

Established Constructs Fail

Performance Expectancy, Effort Expectancy, Facilitating Conditions, Habit: No predictive power for AI. Not just weaker—not significant at all. Challenges core assumptions about technology adoption drivers; may require AI-specific theoretical frameworks.

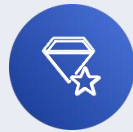
04

Measurement Advancement

First psychometrically validated AI adoption instrument. 8-factor structure with strong psychometrics (CFI = .975). Configural invariance across populations. 85.2% predictive accuracy.

Practical Contributions

Actionable Insights for Organizations



Lead with Value, Not Capabilities

Demonstrate ROI before features. Address total cost: time, disruption, learning curve. Replace feature lists with ROI demonstration. Users are asking "Is this worth it?"



Segment-Specific Strategies

Don't treat all users the same. Four distinct profiles with different needs: 16% enthusiasts (leverage as champions), 30% need proof (provide evidence), 37% follow crowd (show peer adoption), 17% have trust barriers (address before features).



Diagnostic Assessment

3-minute assessment identifies barriers. Targeted interventions vs. one-size-fits-all training. AIRS-16 enables targeted resource allocation instead of blanket programs.



Trust as Prerequisite

17% have trust barriers blocking adoption. Trust-building must precede feature exposure. Feature demonstrations are useless until trust is established. Diagnose trust barriers and address first.

Study Limitations

Study Boundaries

Limitation	Mitigation
Cross-sectional design	Cannot establish causality; shows correlation, not causation. Longitudinal research needed to establish causal relationships.
Self-reported measures	Standard in adoption research but not behavioral; however, behavioral intention validated against actual AI usage ($\rho = .69$)—strong evidence that stated intentions predict actual behavior.
Convenience sample	Through Centiment professional panel; demographics diverse and quality controls rigorous.
US-based	Cultural factors might change relationships; German professionals may respond differently than American counterparts. Cross-cultural replication recommended.
Dropped constructs	4 constructs failed ($\alpha < .60$); redesign needed for Voluntariness, Explainability, Ethical Risk, AI Anxiety.

Constructs That Failed

AI-Specific Measurement Challenges

Four AI-specific constructs failed psychometrically (all reliability below acceptable thresholds). These are measurement challenges requiring better operationalization, not theoretical failures—the constructs matter and require better measurement.

Construct	α	Issue	Future Direction
Voluntariness	.301	Items conceptually conflicted (mandatory vs. choice)	Reframe as "perceived pressure"
Explainability	.503	AI explanation is multi-dimensional; items too broad	Decompose into transparency, interpretability, auditability
Ethical Risk	.582	Near threshold; items conflated personal vs. societal ethics	Separate individual ethical concerns from societal
AI Anxiety	.423	Overlapped with general technology anxiety	Distinguish AI-specific anxiety from general tech anxiety

Future Research: Validation

Recommended Next Steps (Part 1)



Longitudinal Validation

Track actual adoption behavior over time.
Test intention-behavior relationship. Do high behavioral intention scores predict actual adoption at 6 months?



AI-Specific Construct Development

Redesign failed constructs (Explainability, Ethical Risk, AI Anxiety, Voluntariness) with better measurement. Venkatesh (2021) identified nine AI challenges; four addressed here, five remain.



Intervention Testing

Experimental validation of segment-specific strategies. Randomize organizations to targeted vs. generic interventions and measure adoption outcomes. ROI of targeted vs. generic adoption programs.

Future Research: Expansion

Recommended Next Steps (Part 2)

Cross-Cultural Replication

Test measurement invariance across cultures. Examine cultural moderators of adoption drivers. Does Price Value dominate in collectivist cultures? Does Social Influence operate differently? Global validation needed.

Domain-Specific Adaptation

Healthcare, finance, education contexts. Regulated vs. unregulated AI applications. AI in healthcare has different considerations than AI in marketing. The instrument may require domain-specific modules.

Conclusion

Summary of Contributions

Theoretical

- Value-dominant model challenges 30 years of technology acceptance research
- AI Trust as diagnostic extension to UTAUT2
- Evidence that traditional drivers (PE, EE, FC, Habit) fail for AI

Methodological

- AIRS-16: First validated AI adoption readiness instrument
- 8-factor structure with strong psychometrics (CFI = .975)
- 85.2% predictive accuracy

Practical

- 4 distinct user segments with targeted strategies
- Diagnostic tool for organizational assessment
- Evidence-based recommendations for AI implementation

Core insight: Organizations don't have an AI technology problem—they have an AI adoption problem. Understanding and diagnosing that adoption problem is the first step to solving it.

Key Numbers to Remember

523

Sample Size

Diverse participants from
academic and professional
contexts

.505

Strongest Predictor

Price Value β coefficient—
dominates all other factors

.852

Variance Explained

85.2% predictive accuracy—
exceptional model performance

.975

Model Fit (CFI)

Excellent psychometric quality—
exceeds publication standards

4

User Segments

Distinct typologies requiring
different intervention strategies

16

Final Items

Validated instrument length—3-
minute administration time

8

Validated Factors

Constructs with strong reliability
(all $\alpha > .74$)

Thank You

Questions?

Fabio Correa Doctor of Business Administration Candidate
Touro University Worldwide

"Organizations don't have an AI technology problem—they have an AI adoption problem."

Understanding and diagnosing that adoption problem is the first step to solving it. This research provides the diagnostic tools to make that understanding possible.

Backup Slide B1

Full Hypothesis Results Table

H	Path	β	p	95% CI	Result
H1a	PE $\rightarrow$ BI	-.028	.791	[-.234, .178]	X Not Supported
H1b	EE $\rightarrow$ BI	-.008	.875	[-.108, .092]	X Not Supported
H1c	SI $\rightarrow$ BI	.136	.024	[.018, .254]	✓ Supported
H1d	FC $\rightarrow$ BI	.059	.338	[-.062, .180]	X Not Supported
H1e	HM $\rightarrow$ BI	.217	.014	[.044, .390]	✓ Supported
H1f	PV $\rightarrow$ BI	.505	<.001	[.352, .658]	✓ Supported (Strongest)
H1g	HB $\rightarrow$ BI	.023	.631	[-.071, .117]	X Not Supported
H2	TR $\rightarrow$ BI	.106	.064	[-.006, .218]	○ Marginal
H3	Experience mod	.136	.009	—	✓ Partial (HM)
H4	Population mod	-.750	.041	—	✓ Partial (HM)

Complete hypothesis results with confidence intervals. The three supported UTAUT2 hypotheses (Social Influence, Hedonic Motivation, Price Value) all have confidence intervals that do not cross zero. Non-significant paths all have wide intervals crossing zero, confirming lack of effect.

Backup Slide B2

Model Fit Indices Comparison

Index	Obtained	Excellent	Acceptable
CFI	.975	$\geq .95$	$\geq .90$
TLI	.960	$\geq .95$	$\geq .90$
RMSEA	.065	$\leq .06$	$\leq .08$
SRMR	.044	$\leq .08$	$\leq .10$

Source: Hu & Bentler (1999); Browne & Cudeck (1993)

Model fit indices comparison: CFI and TLI exceed excellent thresholds. RMSEA at .065 is in acceptable range, just above .06 excellent threshold—typical for models of this sample size and complexity. SRMR at .044 is excellent. Overall, the measurement model demonstrates very good fit to the data.

Backup Slide B3

What is Price Value?

Price Value measures *perceived worth relative to total investment*, not just licensing costs. Users ask: "Is the total value I'll receive worth the total investment I have to make?"

Direct Costs

Licensing fees, subscriptions, infrastructure investments

Indirect Costs

Learning curve time, workflow disruption, productivity dip during transition

Hidden Costs

Data privacy risks, vendor lock-in, integration complexity

Opportunity Costs

Time spent learning vs. doing current productive work

Price Value evaluation is a richer question than "Is this software expensive?"—it encompasses the full cost-benefit analysis users perform when deciding whether to adopt AI tools.

Backup Slide B4

Why Did Four Constructs Fail?

Construct	α	What Went Wrong	Future Direction
Voluntariness	.301	Items conceptually conflicted (mandatory vs. choice). "My organization requires AI use" and "I choose to use AI voluntarily" interpreted inconsistently.	Reframe as "perceived pressure" to use AI
Explainability	.503	AI explanation is multi-dimensional; items too broad. How model works, why decision made, audit capability are distinct facets.	Decompose into transparency, interpretability, auditability
Ethical Risk	.582	Near threshold; items conflated personal vs. societal ethics. "AI could make biased decisions about me" vs. "AI could harm society"	Separate individual ethical concerns from societal
AI Anxiety	.423	Overlapped with general technology anxiety. Failed to distinguish AI-specific concerns.	Distinguish AI-specific anxiety from general tech anxiety

These are measurement challenges requiring better operationalization, not theoretical failures. The constructs matter and require better measurement approaches.

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